Modeling and Forecasting Natural Gas Prices in India using Generalized Additive Models



Mid-Semester Evaluation Report

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1 Introduction

Natural gas is a pivotal component of India's transition toward a sustainable and diversified energy portfolio. However, the domestic market is highly susceptible to price volatility, driven by fluctuations in global Liquefied Natural Gas (LNG) spot prices and periodic revisions in the government's Administered Pricing Mechanism (APM). Accurate forecasting of Compressed Natural Gas (CNG) and broader natural gas prices is imperative for City Gas Distribution (CGD) entities to plan infrastructure investments and for policymakers to ensure macroeconomic stability.

Traditional time-series and econometric forecasting methods, such as Multiple Linear Regression or ARIMA, assume strictly proportional relationships. These architectures frequently fail in energy markets because they cannot effectively model abrupt regulatory step-changes, non-linear cyclical consumption patterns, and sudden exogenous supply shocks. This project aims to address this critical gap by developing an advanced forecasting framework utilizing Generalized Additive Models (GAMs).

2 Data Architecture and Economic Drivers

A fundamental premise of this research is the distinction between theoretical natural gas *demand* and actual *consumption*. While baseline demand is often projected linearly, actual consumption is dynamically constrained by shifting macroeconomic realities.

The model incorporates variables from distinct domains to construct a Mutually Exclusive, Collectively Exhaustive (MECE) architecture:

- **Sectoral Consumption:** Segmented into transportation usage, proxied by Vehicle Miles Traveled (sourced via the MoRTH Vahan Dashboard), and domestic baseload, represented by Household Utility expansion data (from PNGRB).
- **Macroeconomic Drivers:** Gross Domestic Product (GDP) and Disposable Income metrics (sourced via MoSPI and RBI) to account for purchasing power parity and economic momentum.
- **Dependent Variable:** Historical natural gas pricing data aggregated from the Petroleum Planning and Analysis Cell (PPAC).

3 Methodological Framework

3.1 Generalized Additive Models (GAM)

To overcome the limitations of rigid linear coefficients, the proposed framework employs GAMs. The underlying mathematical structure replaces fixed parameters with smooth, piecewise polynomial functions (splines). The model is defined as:

$$Y_t = \beta_0 + \sum_{i=1}^{p} s_i(X_{it}) + \epsilon_t \tag{1}$$

where Y_t represents the natural gas price at time t, and s_i are smooth functions applied to the independent macroeconomic predictors. This allows the model to naturally calibrate to ascending trends, plateaus, and descents. Furthermore, the model will decompose the time-series into a **polynomial trend** to capture long-term economic trajectories and a **harmonic basis seasonality** to isolate routine cyclical fluctuations.

3.2 Orthogonalization via the Gram-Schmidt Process

A significant econometric challenge in this analysis is the presence of severe multicollinearity. Macroeconomic indicators such as GDP and Disposable Income exhibit high intrinsic correlation, which jeopardizes the statistical validity of the model (a phenomenon known in GAMs as concurvity).

To resolve this, the methodology mandates the transformation of these correlated variables into mathematically independent vectors. This is achieved through the Gram-Schmidt orthogonalization process:

$$\mathbf{u}_k = \mathbf{v}_k - \sum_{j=1}^{k-1} \text{proj}_{\mathbf{u}_j}(\mathbf{v}_k) \tag{2}$$

By ensuring that the inner product of the resulting vectors equals zero ($\langle \mathbf{u}_i, \mathbf{u}_j \rangle = 0$), the macroeconomic drivers become perfectly perpendicular. This mathematical intervention allows the GAM to precisely isolate the distinct impact of each variable without statistical distortion.

4 Current Progress and Future Execution

At this mid-semester evaluation, the project has successfully established the theoretical and mathematical foundations. The critical variables driving non-linear price volatility have been identified, and the necessity of GAMs coupled with Gram-Schmidt orthogonalization has been validated through an extensive review of existing energy forecasting literature.

The subsequent phases of the project will focus on:

1. **Data Normalization:** Aggregating the identified datasets and interpolating mixed-frequency data (e.g., quarterly GDP to monthly metrics).
2. **Algorithmic Implementation:** Programmatically executing the Gram-Schmidt process and constructing the GAM architecture using statistical software (R/Python).
3. **Validation:** Benchmarking the GAM's predictive accuracy (via Root Mean Square Error and Mean Absolute Error) against traditional linear baseline models to quantify the improvement in forecasting reliability.